Rohan Bali

Contact Information

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Links: Website · GitHub · Google Scholar · ORCID

Research Interests

My research is on evaluation methodology and out-of-distribution generalization for spatiotemporal vision. I work on input-design covariates that confound architectural comparisons in earth-observation models (channel count, temporal horizon, geographic distribution), and on whether observed differences hold up under sealed holdouts and zero-shot geographic transfer. Broader interests: robust computer vision, distribution-shift evaluation, calibrated uncertainty in deep learning.

Education

University of Massachusetts Dartmouth, North Dartmouth, Massachusetts, USA

Department of Computer & Information Science

M.S., Data Science, September 2024 – May 2026 | GPA: 3.77/4.0

Advisor: Dr. Donghui Yan

Thesis: *Multi-Horizon Urban Growth Prediction: A Comparative Deep Learning Study*

Focus: Distribution shift, evaluation methodology, and uncertainty quantification in spatiotemporal vision

Manipal University Jaipur, Jaipur, India

School of Computing and Information Technology

B.Tech., Computer and Communication Engineering, August 2017 – May 2021 | GPA: 2.82/4.0

Advisor: Dr. Somya Goyal

Final-year project: *Road Safety Analysis using Machine Learning*

Honors and Awards

GeoAI 2026 Oral Presentation, June 2026

National Talent Search Examination (NTSE) Scholar, Government of India, 2016

Publications

Bali, R. 2026. The Channel-Count Confound: A Continental Audit of Multi-Horizon Urban Growth Prediction. GeoAI 2026, Ghent, Belgium. **(Oral)** Also at SpatialDI 2026, Springer LNCS proceedings, ACM SIGSPATIAL China Chapter, Changsha, China. Sole author. Distilled from the M.S. thesis.

Setup. Audits 5-year vs. 10-year urban-growth prediction across 5,698 CONUS tiles at 250 m under matched evaluation, with a sealed 2020 holdout and zero-shot transfer to 20 Lagos tiles (63,840 pixels).

Channel-count isolation. Moving from 5yr to 10yr removes one epoch, cutting the input stack from 24 to 21 channels. Prior work conflates this covariate with horizon length. 94% of CNN's Figure-of-Merit decline from 5yr to 10yr attributes to the channel reduction (delta-method 95% CI $[62\%, 100\%]$; conservative $\geq$84% floor; channel effect 8× the two-seed $\sigma = 0.007$ spread).

Recency asymmetry. ConvLSTM shows 5.4× larger response to end-removal than to interior-removal. The effect is positional, not volumetric.

Uncertainty and methodology. MC Dropout calibration over 8.69M pixels from 1,352 inference runs yields Pearson $r = 0.983$ between predicted $\hat{\sigma}$ and tile-level error (rank reliable). Absolute magnitude is underestimated by 1.4–1.9×. MSE inverts architectural rankings on land-cover tasks under specific Figure-of-Merit conditions.

GeoAI 2026 short paper (Zenodo). · Code and benchmark repository.

In Preparation

Bali, R. Confident but Unidentifiable: Positivity Limits on Cross-Region Attribution in

Geospatial Models. Sole author. **Transactions in GIS** (Wiley); submitted June 2026. Reframes the question of why a geospatial model fails in a new region as a transportability problem and asks when a covariate-based explanation is even identifiable. Derives a positivity diagnostic and a source-computable partial-identification bound, with a threshold past which not even the sign of an attribution is recoverable. Applied to land-cover transfer from the contiguous United States to 44 cities across four world regions: covariate overlap falls as low as 5.8%, and a confident, well-fit explanation predicts the wrong sign of the transfer gap in every degrading architecture, evidence of a regional effect beyond the measured covariates. Code and data repository.

Bali, R. Train Anywhere, Test Everywhere: Cross-Region Transfer in Earth Observation Is Decided by the Data, Not the Model. Sole author. Prepared as a SIGSPATIAL '26 short paper; follow-on to the channel-count audit. The full source-by-target transfer matrix over 20 world regions shows the training source is inert: the target sets the score (home-field advantage -0.001 FoM, source rows at Spearman $\rho = 0.97$, ordering identical across six architectures incl. fine-tuned foundation models), and a parameter-free recent-past baseline tops every trained model (FoM 0.56 vs. 0.34, IoU 0.92 vs. 0.61). Swapping the input traces a provenance spectrum, from harmonised products that transfer to raw sensor reflectance that does not; the same target-over-model dominance recurs on the PANGAEA benchmark. Code and data repository.

Technical Methods

Spatiotemporal architectures: ConvLSTM, CNN, U-Net, sequence models (LSTM, GRU, Transformer) for forecasting.

Evaluation under distribution shift: matched-tile audits, sealed temporal holdouts, zero-shot geographic transfer, Figure-of-Merit, MSE/FoM rank-inversion analysis, channel-volume controls.

Uncertainty quantification: MC Dropout calibration at pixel scale, reliability diagrams, predictive variance under horizon shift.

Earth-observation data: Landsat, Sentinel, NLCD; continental-scale tiling, harmonization, and export on Google Earth Engine.

Stack: Python, PyTorch, TensorFlow, xarray, Rasterio, GeoPandas, GDAL, scikit-learn, NumPy, Pandas.

Selected Professional Experience

Capgemini, Mumbai, India

Software Engineer **December 2021 – August 2023**

Supported production infrastructure and job-scheduling workflows; built Python, PowerShell, and Bash automation for data and systems operations across cloud environments.

Upcred.ai, Bangalore, India

Machine Learning Researcher **June 2021 – December 2021**

Designed Python and Beautiful Soup web-scraping pipelines and MongoDB schemas supplying training data for the team's machine learning models.

Service and Volunteering

Founding Team Member, The Parivartan Foundation **2023 – Present**

Registered non-profit based in Shimla, Himachal Pradesh, India. Source and analyze public government datasets to identify community-level priorities and inform targeted outreach campaigns for local development and civic engagement.